

ROBOTIC ARM

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Problem statement

- **Predicting Robot Arm Accuracy in Pick-and-Place Operations**
In modern automated manufacturing systems, robotic arms play a crucial role in performing precise pick-and-place operations. However, deviations from the target location during these operations can lead to quality defects, reduced efficiency, and increased operational costs. This project aims to develop a regression model that can accurately predict the deviation from the target location of a robot arm based on various operational and environmental parameters.

Datasets used

- **Joint angle, Arm length, Payload, Speed, Torque, Task complexity, Calibration frequency, Controller type, Workspace temp, Cycle count. Algorithms used**

Algorithms used

Linear regression

Random forest

Decision tree

Support vector regression

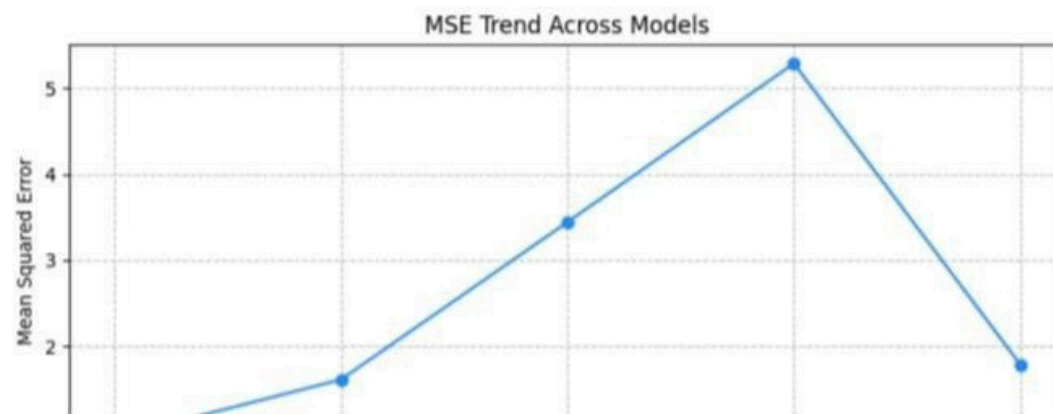
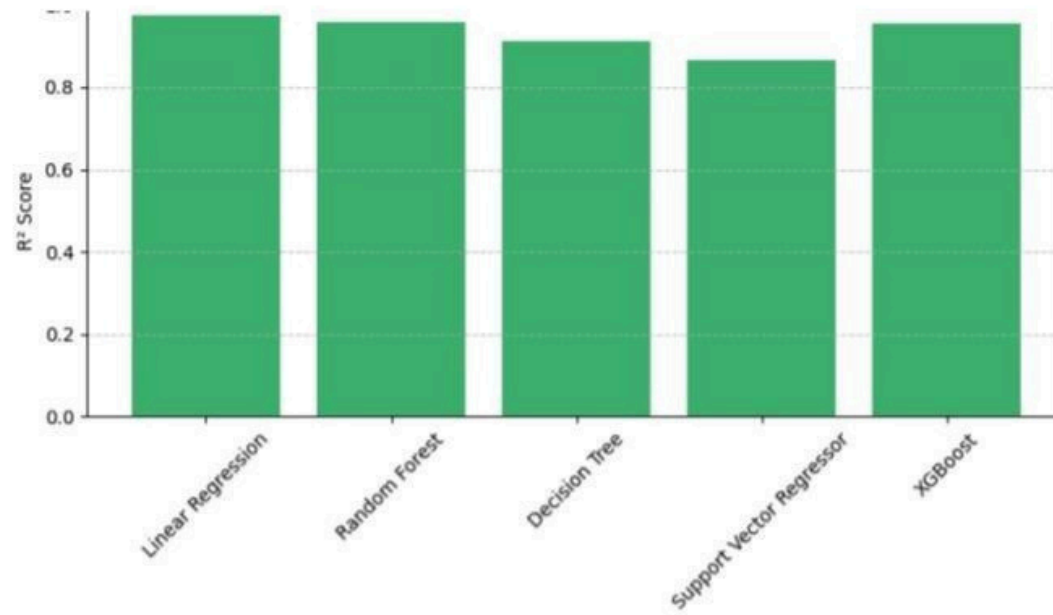
XGBoost

Why Linear Regression is Best

- - Linear Regression is a powerful tool for predicting outcomes based on sensor input (e.g., position, force)
- - It's simple, fast, and interpretable, making it ideal for real-time control and error prediction
- - Helps in calibrating motion paths, estimating torque, and predictive maintenance
- Conclusion: Linear regression is best for its balance of accuracy, efficiency, and ease of implementation in robotic systems.

BENIFITS:

- Improves precision and reduces errors**
- Increases efficiency and productivity**
- Enables faster and more reliable operation**
- Minimizes downtime and repairs through predictive maintenance**
- Enhances quality control and detects errors**
 - Optimizes robot performance**
 - Applicable in various industries such as manufacturing, logistics, food processing, and pharmaceuticals**



Model Contribution Based on R^2 Score

